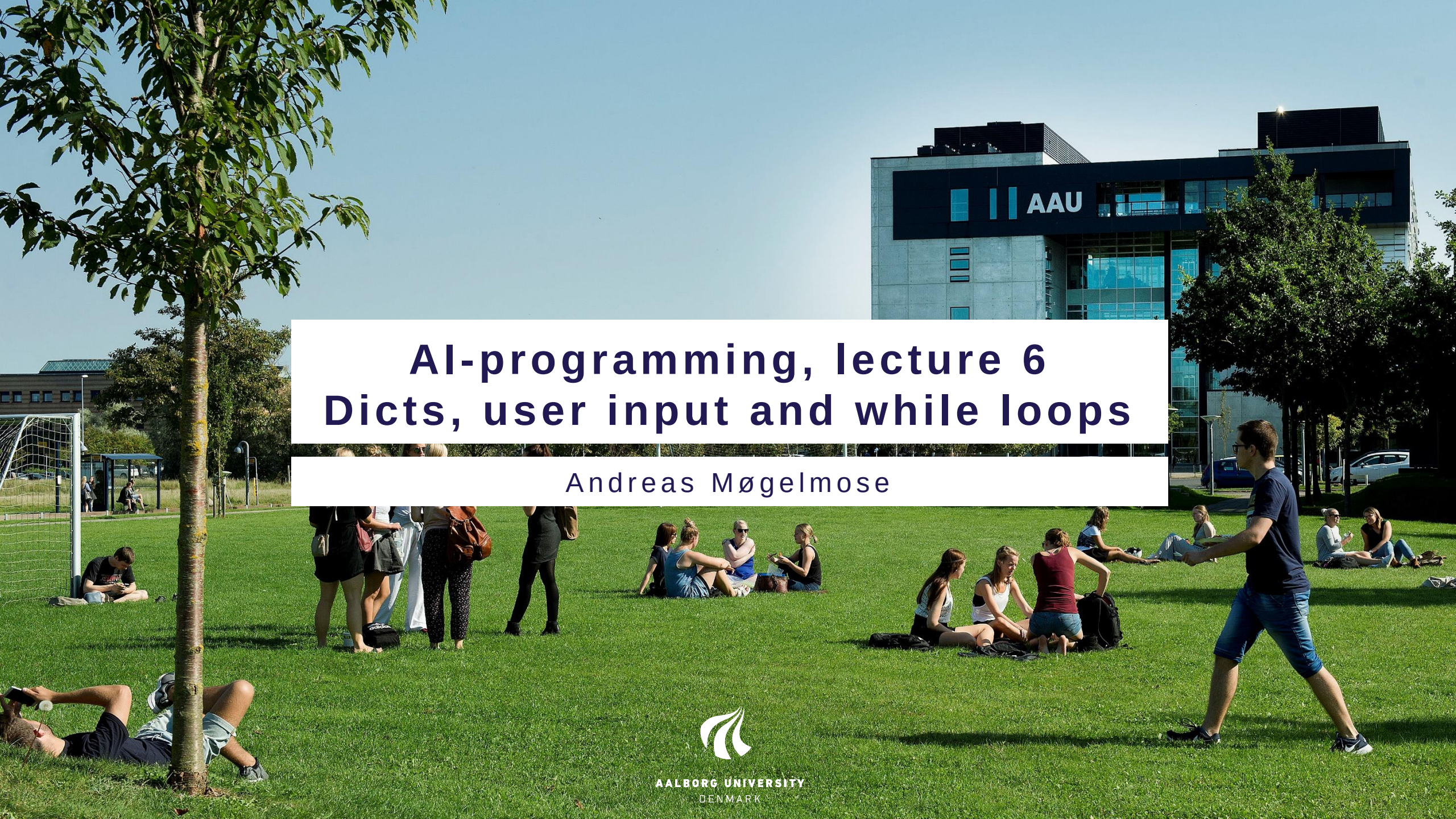


AI-programming, lecture 6
Dicts, user input and while loops

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Agenda

- Dictionaries (and sets)
- User input
- While loops



Recap: Functions

```
my_string.lower()

move(7, 2)

QuitProgram()
```



Recap: Variables and lists

```
name = 'Bob'
age = 21
print(f"My name is {name}. I am {age} years old.")

print(20 % 7)

vegetables = ['carrot', 'lettuce', 'onion', 'radish']
vegetables.append('potato')
```



Recap: Loops

```
cars = ['audi', 'bmw', 'subaru', 'toyota']
cars = [car.title() for car in cars]

for car in cars[1:3]:
    print(car)
```



Recap: Branching

```
while True:
    screen.fill((100, 200, 100))
    s = datetime.now().second
    if s > 30 and enable_rect:
        pygame.draw.rect(screen, (200, 100, 100), (30, 70, 140, 60))
        if s == 59:
            enable_rect = False
    else:
        pygame.draw.ellipse(screen, (100, 100, 200), (30, 70, 140, 60))
```



Agenda

- **Dictionaries (and sets)**
- User input
- While loops



Dictionary basics

```
car = {"brand": "Ford", "model": "Mustang", "year": 1964}
print(f"I have a {car['brand']} {car['model']} from {car['year']}")

car["year"] = 2020
print(f"Oops, it's actually from {car['year']}")

car["color"] = "red"
print(f"The car is {car['color']}")
```



Looping over dictionaries

```
birthdays = {"Alice": "Apr 1",
             "Bob": "Dec 12",
             "Carol": "Mar 4"}

for name, date in birthdays.items():
    print(f"{name} was born on {date}")

for name in birthdays:
    print(name)

if "Andreas" not in birthdays.keys():
    print("Andreas' birthday is not known :-(")
```

Dicts in lists and lists in dicts

```
my_cars = [{"brand": "Ford", "model": "Mustang", "year": 1964},
           {"brand": "Tesla", "model": "Model S", "year": 2019},
           {"brand": "Volkswagen", "model": "Golf", "year": 2014}]

other_peoples_cars = {"Alice": ["Ford", "GMC"],
                      "Bob": ["Tesla", "Volkswagen"]}

other_peoples_cars["Alice"].append("Toyota")
```



Sets

```
>>> ages = [13, 15, 5, 32, 6, 5, 13]
>>> ages
[13, 15, 5, 32, 6, 5, 13]
>>>
>>> ages = set([13, 15, 5, 32, 6, 5, 13])
>>> ages
{32, 5, 6, 13, 15}
```

```
>>> ages = {13, 15, 5, 32, 6, 5, 13}
>>> ages
{32, 5, 6, 13, 15}
```



Sets

```
>>> programming_students = {"Alice", "Bob", "Charlie"}
>>> rock_climbers = {"Rick", "Charlie", "Daniel"}
>>> programming_students.union(rock_climbers)
{'Bob', 'Alice', 'Charlie', 'Daniel', 'Rick'}
>>> programming_students.intersection(rock_climbers)
{'Charlie'}
```

- difference()
- symmetric_difference()

- isdisjoint()
- issubset()
- issuperset()



Agenda

- Dictionaries (and sets)
- **User input**
- While loops



Basic user input

```
name = input("What is your name? ")
print(f"Hello, {name}!")

age = int(input("How old are you? "))
print(f"You are {age} years old")
```

```
What is your name? Jens
Hello, Jens!
How old are you? 22
You are 22 years old
```



Pygame user input

- event.type:
  - KEYDOWN: Tastatur (key, mod, unicode)
  - KEYUP: Tastatur (key, mod)
  - MOUSEBUTTONUP: Mus (pos, button)
  - MOUSEBUTTONDOWN: Mus (pos, button)
  - MOUSEMOTION: Mus (pos, rel, buttons)

- pygame.K_LEFT
- pygame.K_a
- https://www.pygame.org/docs/ref/key.html#key-constants-label

Examples

```
import pygame
import math
from datetime import datetime

pygame.init()
screen = pygame.display.set_mode((640, 480))

box_position = (100, 100)
box_size = (25, 25)

while True:
    screen.fill((255, 255, 255))
    rect = box_position + box_size
    pygame.draw.rect(screen, (0,0,0), rect)
    pygame.display.flip()

    for event in pygame.event.get():
        if event.type == pygame.QUIT:
            pygame.quit()
            exit()
        if event.type == pygame.KEYDOWN:
            if event.key == pygame.K_LEFT:
                box_position = (box_position[0] - 5, box_position[1])
            if event.key == pygame.K_RIGHT:
                box_position = (box_position[0] + 5, box_position[1])
            if event.key == pygame.K_UP:
                box_position = (box_position[0], box_position[1] - 5)
            if event.key == pygame.K_DOWN:
                box_position = (box_position[0], box_position[1] + 5)
```

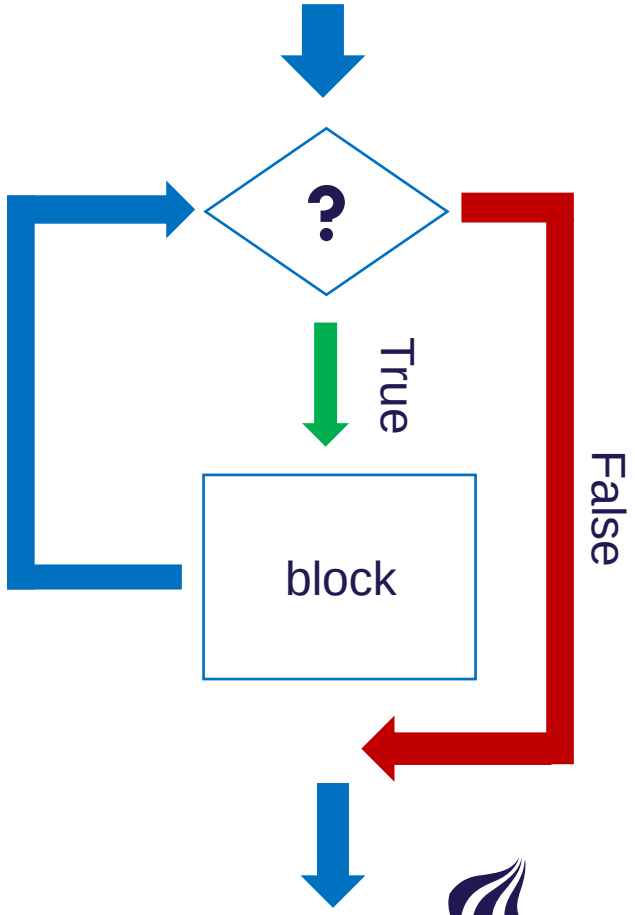
Agenda

- Dictionaries (and sets)
- User input
- **While loops**



Condition-based loops

- The `while` loop



Condition-based loops

- The `while` loop
- Evaluates the condition *before* each iteration
- Equivalent to repeating an `if` statement

```
code before
while condition:
    code block
code after
```



Condition-based loops

- Terminating the `while` loop
  - The 3 typical usages
  - The infinite loop

```
while (true)
{
    code block
}
```

  - Termination on external event

```
while (!event)
{
    code block
}
```

  - Termination based on variable

```
while (condition w. variable)
{
    code block that
    changes the variable
}
```



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while examples

```
while True:
    for event in pygame.event.get():
        if event.type == pygame.QUIT:
            pygame.quit()
            exit()
```

```
run_flag = True
while run_flag:
    for event in pygame.event.get():
        if event.type == pygame.QUIT:
            run_flag = False
```



break

```
i = 0
while i < 10:
    if i == 5:
        break
    print(i)
    i += 1
```

```
cars = ["Ford",
        "GMC",
        "Tesla",
        "Volkswagen",
        "Toyota"]

for car in cars:
    if car == "Tesla":
        break
    print(car)
```

continue

```
i = 0
while i < 10:
    if i == 5:
        continue
    print(i)
    i += 1
```

```
cars = ["Ford",
        "GMC",
        "Tesla",
        "Volkswagen",
        "Toyota"]

for car in cars:
    if car == "Tesla":
        continue
    print(car)
```



while for list processing

```
cars = ["Ford",
        "GMC",
        "Tesla",
        "Volkswagen",
        "Toyota"]

print(cars)

while len(cars) > 0:
    print(cars.pop())

print(cars)
```



Exercise time!